
Regression Challenge

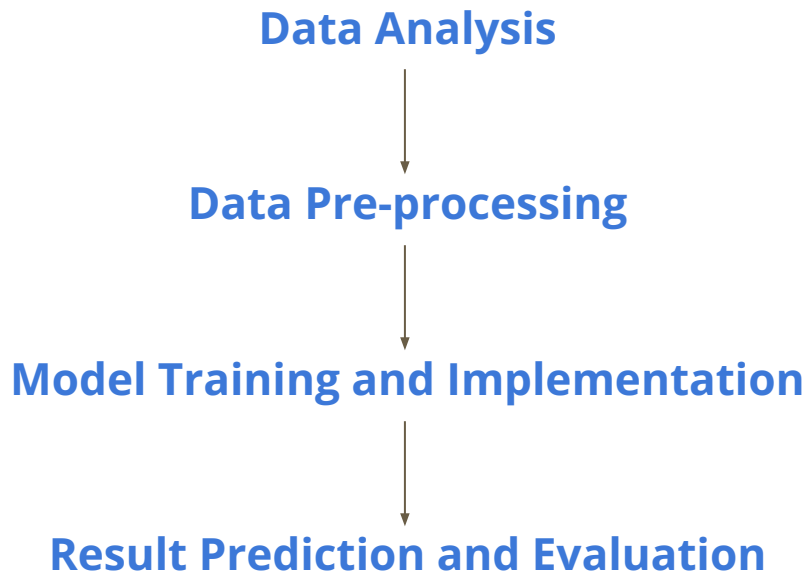
Faisal Farhan

Erasmus Mundus Joint Masters Degree in Medical Imaging and Applications (MAIA)
Statistical Learning and Data Mining
Università degli studi di Cassino e del Lazio Meridionale, Italy

Regression Problem using parametric and
Non parametric approach



Methodology



1

Checking dataset, correlation matrix, identifying outliers, linearity, non-linearity of the dataset.

2

Removing irrelevant columns, outliers, splitting dataset to validation and train set for evaluation

3

Fitting the data with two regression models- Linear regression and KNN. Addressing potential problems like collinearity, non constant variance et.

4

Evaluating the model on validation and train data and prediction of results on new test dataset.

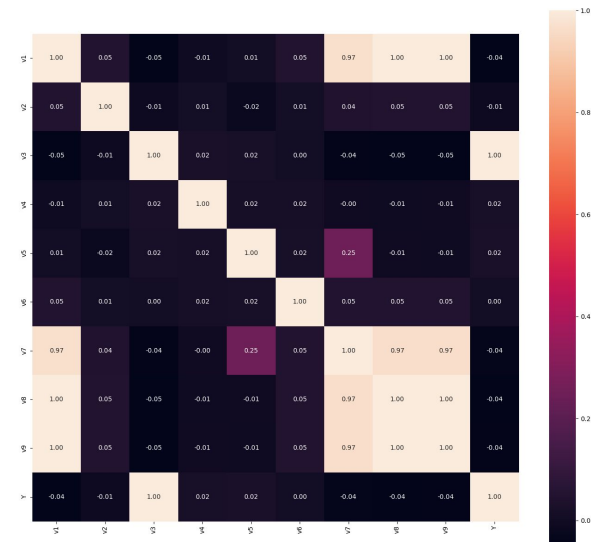
Data Analysis

Correlation Matrix:

v1 shows a strong positive correlation with v8 and v9, which might be close to 1.

v2 seems to have little to no correlation with most of the other variables, as indicated by the lighter colors.

The strong correlations between v1, v8, and v9 could indicate multicollinearity.



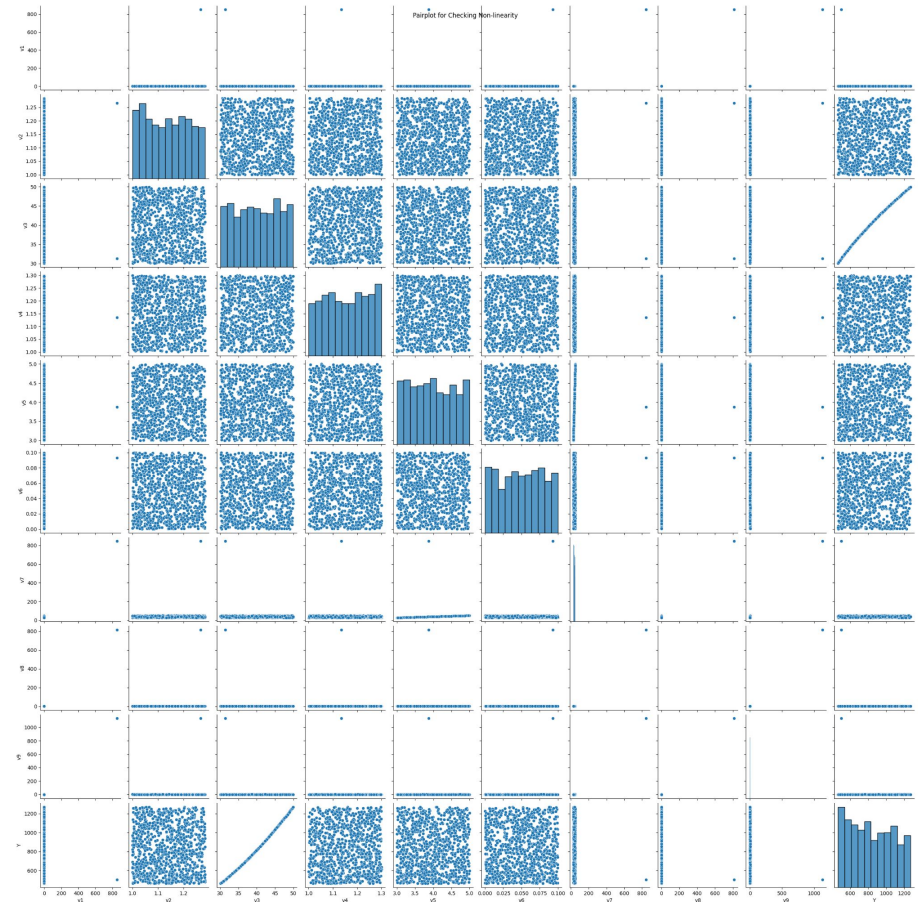
Data Analysis

Linear Non-linear relationship:

-Most variables appear to have a fairly uniform distribution, though some skewness can be observed in certain variables. For instance, v1, v2, v4, v5, and v6 show a left-skewed distribution, while v3, v7, v8, and Y show a right-skewed distribution.

-The scatter plots do not show distinct linear relationships between the independent variables and the target variable Y. Instead, the points appear quite scattered, which suggests that linear relationships, if present, are not strong or may be obscured by the scale of the data.

-There are some potential outliers visible in the scatter plots involving the target variable Y, particularly at higher values. These outliers could have a significant impact on the fit of a regression model



Data Analysis

Interaction Terms:

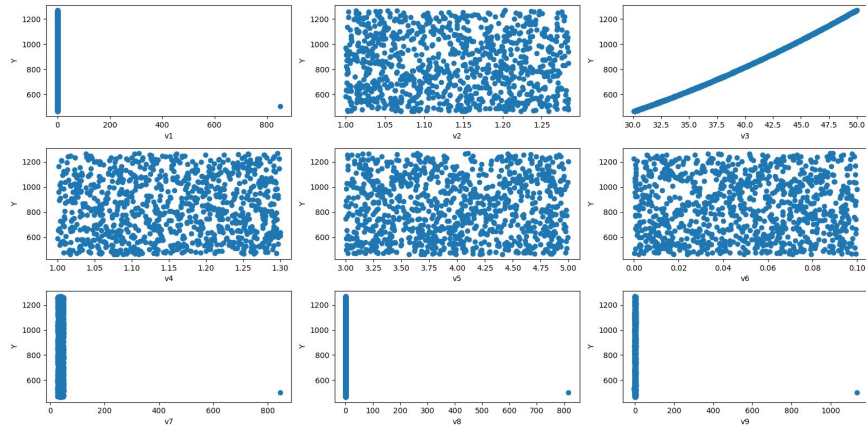
	coef	std err	t	P> t	[0.025	0.975]
x1	9.7322	2219.652	0.004	0.997	-4347.382	4366.846
x2	-0.4079	29.811	-0.014	0.989	-58.927	58.111
x3	236.6850	29.774	7.949	0.000	178.240	295.130
x4	0.1113	29.779	0.004	0.997	-58.344	58.566
x5	18.1010	8609.405	0.002	0.998	-1.69e+04	1.69e+04
x6	-0.4676	29.855	-0.016	0.988	-59.071	58.136
x7	-26.9137	9836.096	-0.003	0.998	-1.93e+04	1.93e+04
x8	0.3574	29.866	0.012	0.990	-58.269	58.984
x9	0.4844	29.886	0.016	0.987	-58.180	59.149
=====						
Omnibus:		145.561	Durbin-Watson:			0.001
Prob(Omnibus):		0.000	Jarque-Bera (JB):			74.293
Skew:		0.595	Prob(JB):			7.37e-17
Kurtosis:		2.097	Cond. No.			763.
=====						

Linear Regression

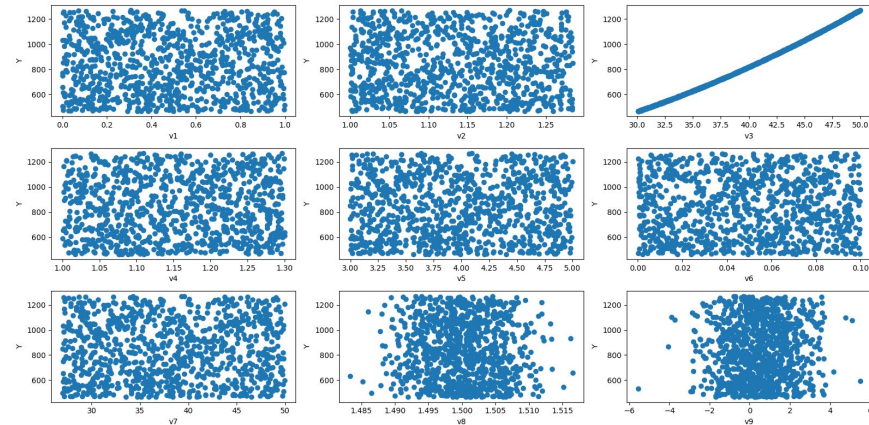
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KNN

Data Pre-processing



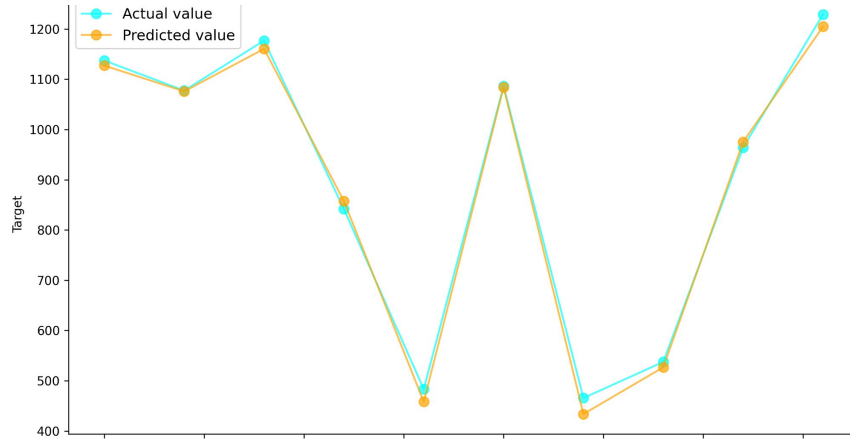
Scatter plot with outliers



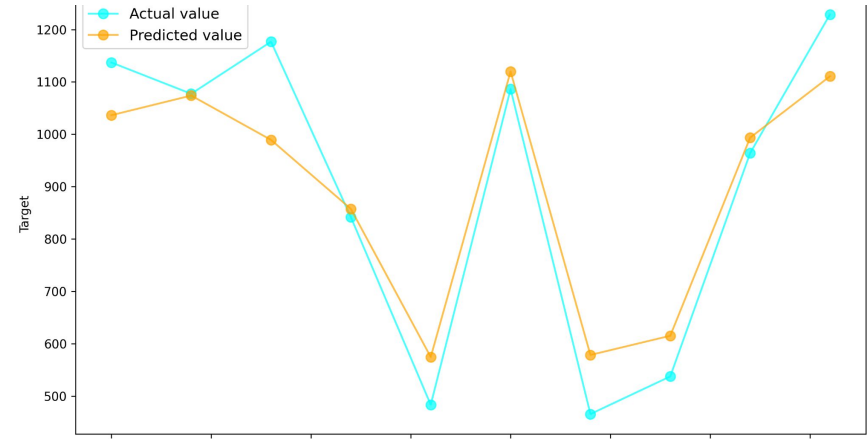
Scatter plot after removing outliers

- Unnamed column indicating index are removed.
- Outliers are identified and removed based on Z-score.
- Data is split into training and validation set for further evaluation.
- Training and validation data are standardized to ensure that each feature contributes equally to the distance calculations.

Model Training and Implementation



Actual vs Predicted value on validation set (Linear Regression)



Actual vs Predicted value on validation set (KNN)

- Linear Regression model and KNN model is fit on the dataset.
- Model is first evaluated with training and validation set.
- Actual and Predicted values for validation set is measured.
- Performance metrics like Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R2 score is also calculated for training and validation set.

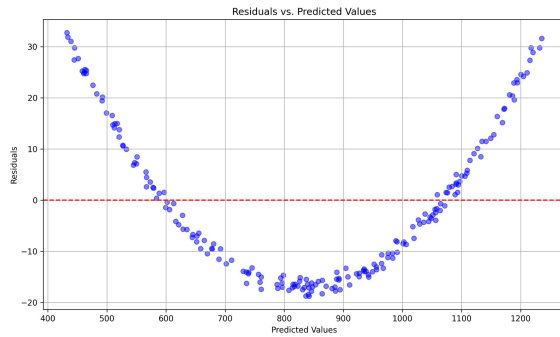
Model Training and Implementation

Model Name	Training Set			Validation Set		
	R2	MAE	RMSE	R2	MAE	RMSE
Linear Regression	0.996	12.802	14.856	0.995	12.64	14.762
KNN	0.928	50.961	63.272	0.895	60.462	74.166

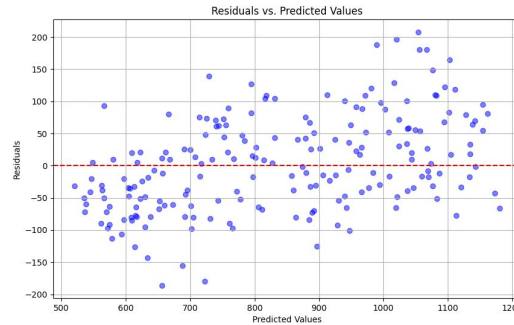
- Based on the performance evaluation metrics it can be concluded that Linear Regression model outperforms KNN model and can provide more accurate values on the given dataset.

Model Training and Implementation

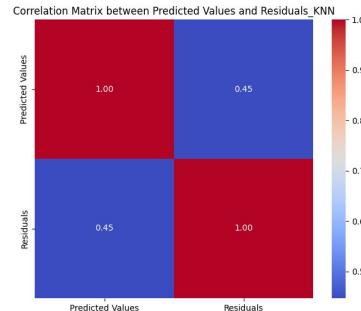
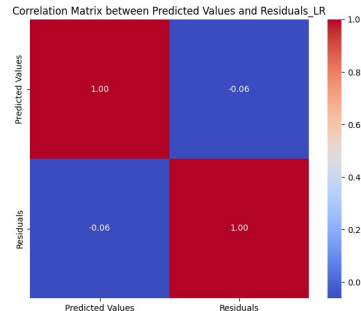
Residual Plots and Correlation of Error Terms:



Residual plot Linear Regression



Residual plot KNN



- The residual plots show that they exhibit non-linearity. It is more evident in the plot of Linear Regression.
- Based on the correlation of error terms we can say that, the linear regression model seems to be more effective than the KNN model for this particular dataset in terms of error randomness and the independence of residuals, which is an assumption of the linear regression model.
- For the KNN model, the presence of a moderate correlation suggests that the KNN might be systematically overestimating or underestimating for certain regions of the feature space.

Result Prediction and Analysis

- The trained models (Linear Regression and KNN) are then implemented on the given test_ch.csv dataset for making predictions of the target variable.
- The first column, contain the output values estimated using linear regression, the second the estimates obtained with KNN.

	Linear_Regression	KNN
0	9204.840899	772.513532
1	7466.874503	527.106479
2	11398.992488	1148.194534
3	7702.065662	561.608138
4	10760.550089	1084.394159
5	7986.024976	559.195148
6	8966.639824	864.668073
7	7754.860568	660.561757
8	10455.371631	962.252286
9	10584.461374	1030.009785